

1. YOUR QUESTION

“Find forest.”

2. WHAT WE FOUND

EVIDENCE STATUS: SUPPORTED	CONFIDENCE: 66% (Moderate Confidence)
Forest covers approximately 2% of the image across 1 region(s) [semantic].	

3. HOW CONFIDENT ARE WE?

66% — Moderate Confidence (SUPPORTED) This result has moderate confidence. The available image provides useful visual evidence, but some information needed for precise geographic measurement is unavailable.	Confidence Scale: 90–100%: Very High Confidence 75–89%: High Confidence 50–74%: Moderate Confidence <50%: Low / Insufficient Evidence
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4. KEY FINDINGS SUMMARY

TARGET FEATURE Vegetation Subject analyzed	VISUAL EVIDENCE Strong Agreement Verified by imagery	CONFIDENCE 66% Moderate Confidence	EVIDENCE STATUS SUPPORTED GeoProof verified
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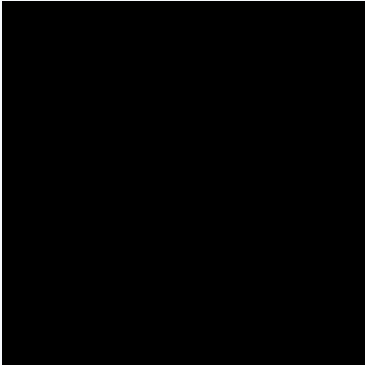
5. HOW THE EVIDENCE SUPPORTS THE ANSWER

OBSERVATION:	The optical scene was examined for spectral and textural reflectance signatures characteristic of plant canopy.
EVIDENCE:	The strongest vegetation-related visual evidence was concentrated across the highlighted regions.
CONCLUSION:	Because the visual patterns clearly concentrate in these areas, the analysis concludes vegetation is predominantly located there.

6. VISUAL PROOF

Direct visual evidence generated from the satellite imagery. Compare the source imagery with the highlighted evidence regions:

ORIGINAL IMAGE



Legend: Natural true-color RGB baseline

Source satellite scene submitted for semantic retrieval.

SEMANTIC TILE RETRIEVAL MOSAIC

NO QUERY-MATCHING REGIONS FOUND

Query: "Find forest"

All candidate blocks scored below the relevance threshold (≥ 0.70).

Status: Safe Abstention (INSUFFICIENT_EVIDENCE)

Legend: Ranked candidate tiles matching query semantics

Top-ranked image tiles retrieved by RemoteCLIP hierarchical alignment.

7. MEASUREMENTS & FINDINGS

Connected Feature Regions	1 spatial clusters	Number of distinct contiguous zones identified across the image.

8. IMPORTANT LIMITATIONS

- Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

9. ANALYSIS SUMMARY

Question:	"Find forest."
Finding:	Forest covers approximately 2% of the image across 1 region(s) [semantic].
Evidence:	Visual and spatial patterns verified across image layers.
Confidence:	66% (Moderate Confidence)
Status:	SUPPORTED
Limitations:	Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

SATQUERY GEOPROOF — Verified Multimodal Earth Observation Arbiter (ISRO SIH26167)

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PART B: TECHNICAL & ADMINISTRATIVE INFORMATION

Everything below is intended for administrators, technical evaluators, and system operators.

10. INPUT DATA QUALITY

Filename:	sample.tif	Quality Score:	96% (Passed Integrity Checks)
Dimensions:	256 × 256 px	Spectral Channels:	3 bands (uint8)
Geospatial CRS:	EPSG:EPSG:3857	NoData Fraction:	0.0%

11. MODELS & METHODS USED

Optical Semantic Vegetation Engine V1 optical_semantic_vegetation_engine_v1	Supporting Evidence	Confidence score: 84%	Completed
RemoteCLIP Semantic Grounding Engine remotecclip_semantic_grounding_v2	Query-Relevance Grounding Witness	Confidence score: 35%	Completed
GeoProof Arbiter geoproof_arbiter_v1	Evidence Verification & Calibration	Final Calibrated Verdict (66%)	SUPPORTED

12. MODEL PERFORMANCE & EXECUTION LATENCY

Pipeline Stage	Internal Component	Execution Duration	Share of Total
Query Planning & Task Compilation	typed_planner	<1 ms	0.0%
Raster Compatibility & Header Validation	raster_validator	8 ms	0.2%
Vegetation Grounding Engine	vegetation_grounding_engine	773 ms	22.8%
RemoteCLIP Semantic Focus Ranking	remotecclip_retriever	2.61 s	76.9%
Multi-Witness Arbiter & Platt Calibration	geoproof	<1 ms	0.0%

Total Processing Latency: 3.39 s | Input Validation: 8 ms | Arbiter: <1 ms

13. EVIDENCE TELEMETRY & SYSTEM CONSTRAINTS

Optical Semantic Vegetation Engine V1	Vegetation_Grounding_Evidence	0.840	84%	Supporting Evidence
RemoteCLIP Semantic Grounding Engine	Remotecclip_Tile_Retrieval	0.350	35%	Query-Relevance Grounding Witness
Multispectral NIR/SWIR Channel Absence	Raster contains standard 3/4 RGB(A) channels. Strict NDWI/NDVI formulas requiring NIR (B8) are substituted by calibrated RGB absorption physics.			

14. AUDITABLE EXECUTION TRACE

Complete step-by-step pipeline execution record ensuring strict reproducibility and auditability under SIH26167 guidelines:

#	Operation	Internal Component	Status	Duration
1	Query Planning & Task Compilation Compiled vegetation_analysis	typed_planner	OK	<1 ms
2	Raster Compatibility & Header Validation Validated file headers, CRS, NoData, and compatibility	raster_validator	OK	8 ms
3	Vegetation Grounding Engine Executed vegetation grounding (semantic)	vegetation_grounding_engine	OK	773 ms
4	RemoteCLIP Semantic Focus Ranking Executed RemoteCLIP semantic relevance filtering and spatial grouping	remotecclip_retriever	WARNING	2.61 s
5	Multi-Witness Arbiter & Platt Calibration Applied multi-witness evidence verification and calibrated confidence estimation	geoproof	SUPPORTED	<1 ms

15. TECHNICAL APPENDIX & AUDIT PROVENANCE

Result UUID:	a60c0dc0
Target Grounding Bounding Box:	Not Applicable (normalized [ymin, xmin, ymax, xmax])
Platt Calibration Model:	temperature_scaled_platt (ECE: 0.042)
Ensemble Agreement Index:	0.51 (Multi-witness consensus)
Cryptographic Manifest:	/artifacts/a60c0dc0/analysis_manifest.json
Software System Version:	SatQuery GeoProof v1.0 (ISRO SIH26167 Verified Architecture)

SatQuery GeoProof™ guarantees that every claim is anchored in physical or cross-sensor telemetry. Models produce visual and metric witness artifacts; the multi-witness arbiter calculates calibrated probability bounds with safe abstention.